Appendix

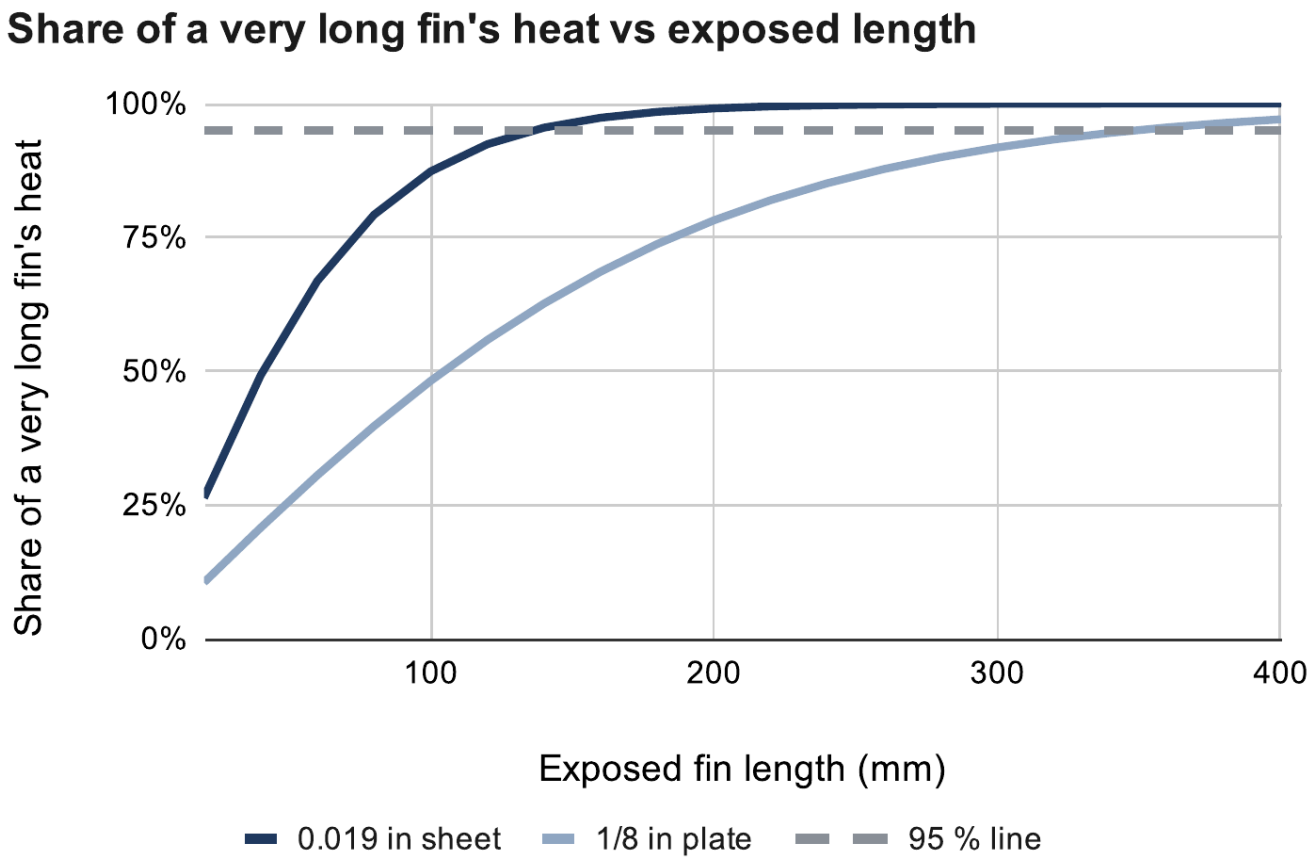
Fundamentals, safe state, fits, schematics

Safe state and failure modes

Event	What happens	Mitigation
PAUSE, or a fault the software detects	Every station holds position with torque on	Nothing moves again without approval
Wi-Fi link lost	Stations stop and hold on their own clocks	Resume needs approval
Power loss or emergency stop	Torque is lost and the claw can fall (1.5 kg from about 2 m)	Nobody underneath; claw parked low when idle; brake planned
Controller reset or brown-out	Driver disabled at boot, so torque is lost	Same as power loss
Driver over-temperature	The A4988 shuts its outputs off	Hold test watches the driver; fallback 1.35 A
Three-station run: any station fault	Every cable is needed; the claw falls	Floor clear and padded; movement and calibration only

Heat fin length

CALCULATED



- 0.019 in prototype reaches 95 % of its capacity by 136 mm
- 1/8 in design keeps gaining to 349 mm
- Built at 200 mm, limited by ceiling clearance

Calculated (calc sheet: Heat fin)

Fin length and efficiency

$m = \sqrt{(2h / (k\,t))}$, $L_{95} = 1.83 / m$

$\eta = \tanh(mL) / (mL) = \mathbf{0.74}$ (1/8 in)

Fin sizing for 4 W

CALCULATED

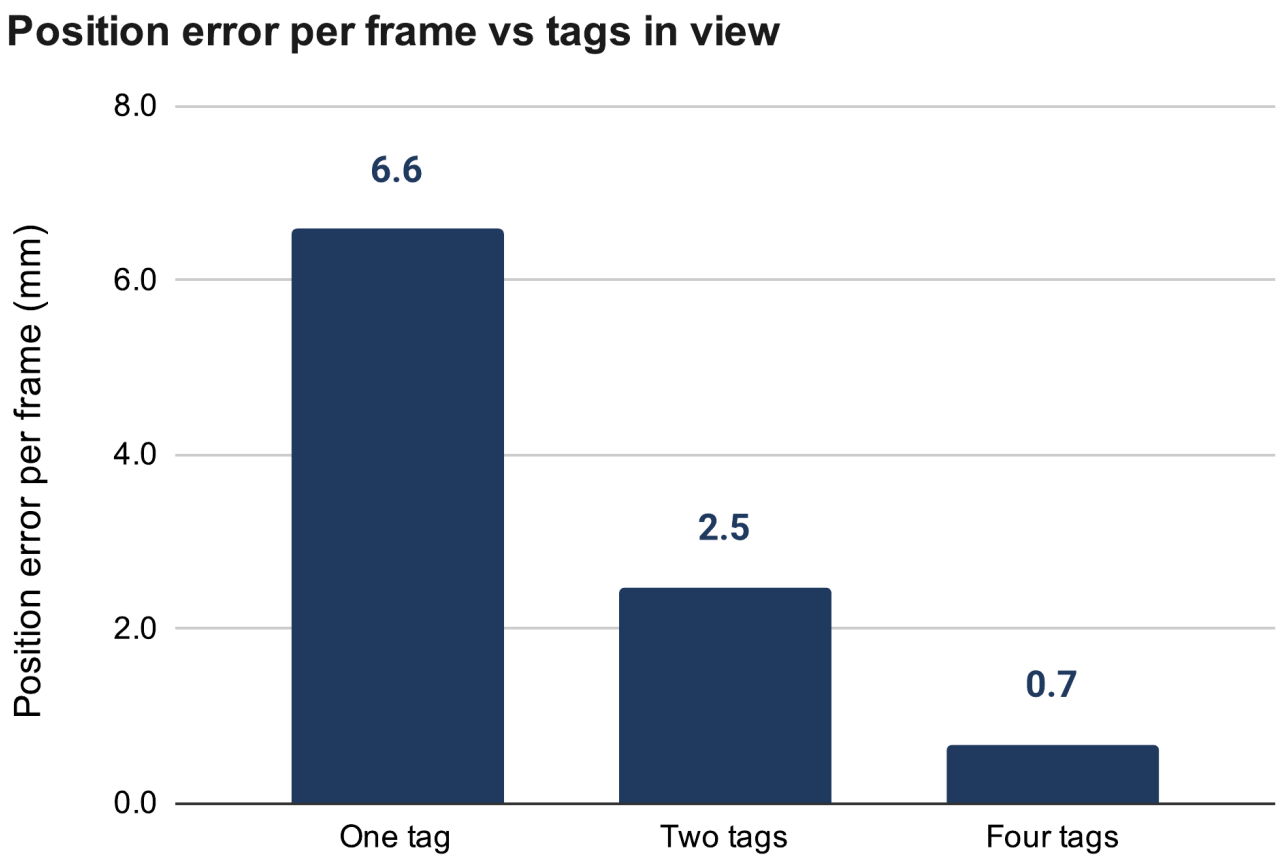
Painted 6061, 44 mm wide	Mass	Shortest for ≤ 60 °C
0.019 in (prototype)	14 g	None (2.0 W max)
1/16 in	47 g	None (3.6 W max)
0.090 in	68 g	210 mm or none
1/8 in (design)	94 g	145–185 mm
3/16 in	141 g	125–140 mm
1/4 in	189 g	115–125 mm

All 4.05 W through the fin, 23 °C room, paste contact. Calculated.

- If the fin alone had to carry all 4.05 W
- 0.019 in sheet: not possible at any length
- 1/8 in plate: 145–185 mm

Floor board accuracy

SIMULATED

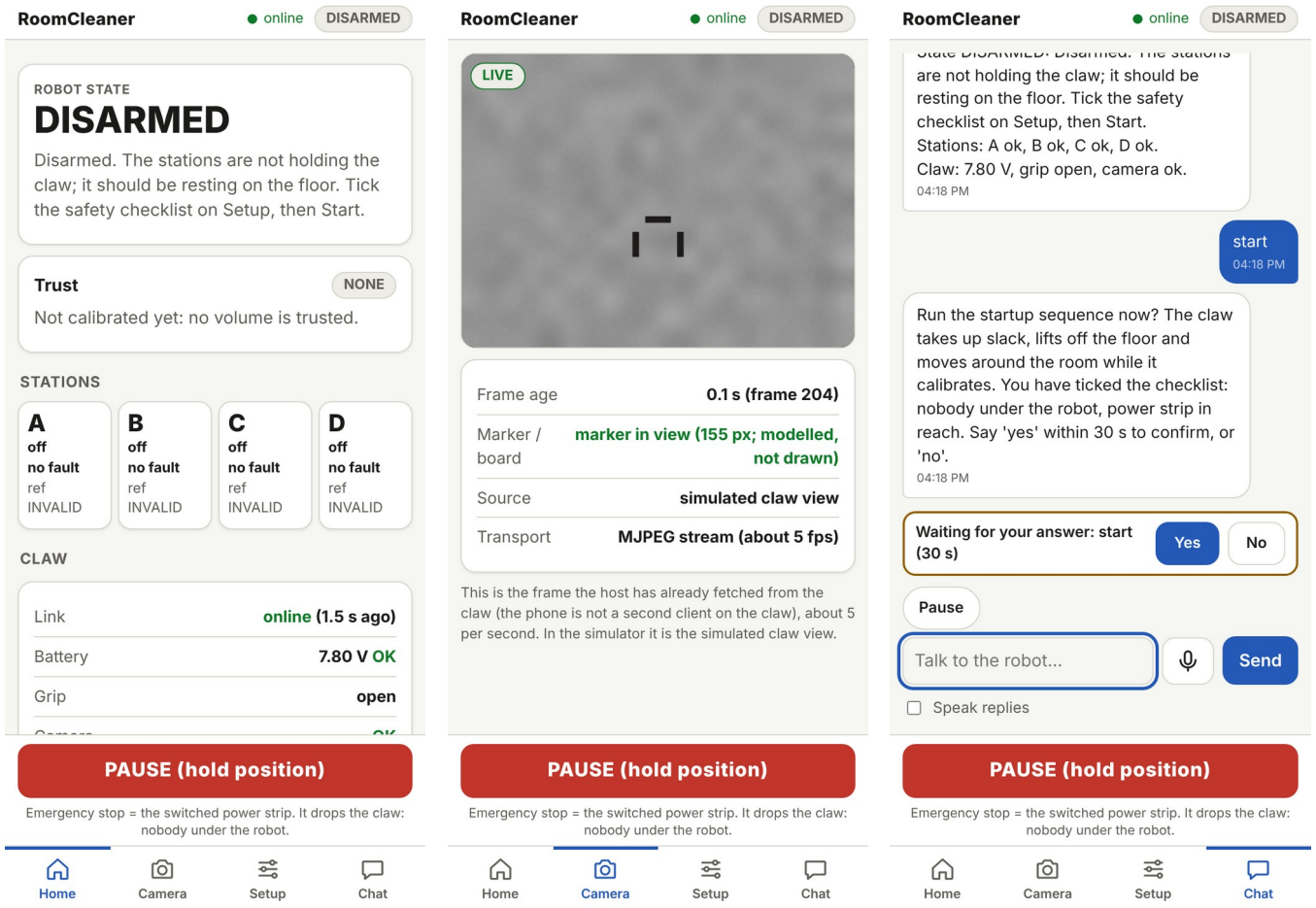


- Camera pose error per frame, best case: 6.6 mm with one tag, 0.7 mm with four
- This is the camera's view, not the cable length
- After the fit, cable lengths are known to about 12–16 mm (simulated)

Calculated (calc sheet: Floor board)

Operating sequence

SIMULATED



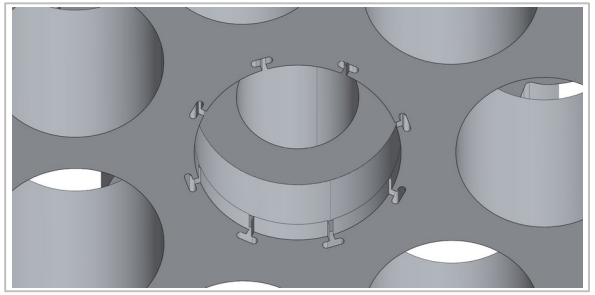
Phone app in simulation: home, claw camera and chat

- Start up: self-check, take up slack, calibrate
- Find: scan from above, detect garments, choose the nearest
- Pick: hover until still, grip, deliver; the camera confirms each step
- Every move waits for approval on the phone; PAUSE holds position

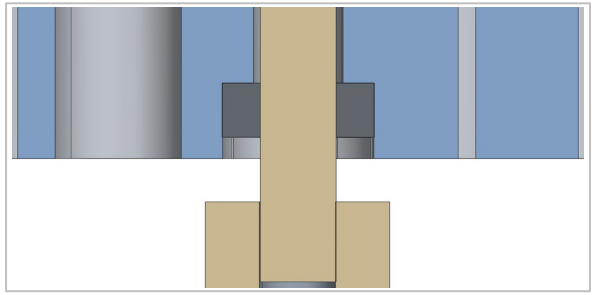
Control logic: Generation 1 vs. final

	Generation 1	Final
Where is the claw?	Assumed from the commanded cable lengths	Measured by the claw camera; refit at every startup
Room geometry	Anchors from a tape measure	Anchors and cable lengths fitted, with error bars
Motor control	One central board	A controller per station, coordinated by the laptop
Finding laundry	Ceiling camera	Claw camera
Did the grab work?	Assumed	Camera checks pickup and delivery
Faults	Emergency stop only	PAUSE keeps torque; the emergency stop cuts power and drops the claw

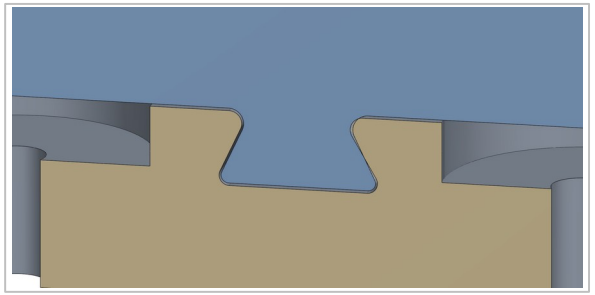
Manufacturing: fits and datums



Gear: Ø14.000 entry lip, 8 T-slots



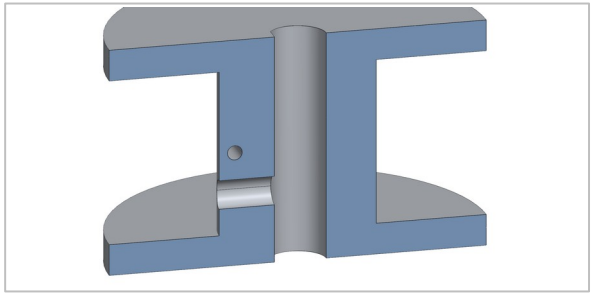
Cut open: 687 bearing on the printed pin



Slide-in rib in the wall-mount dovetail



End-stop switch pressed into the slide-in



v3 spool: D-bore and set-screw hole



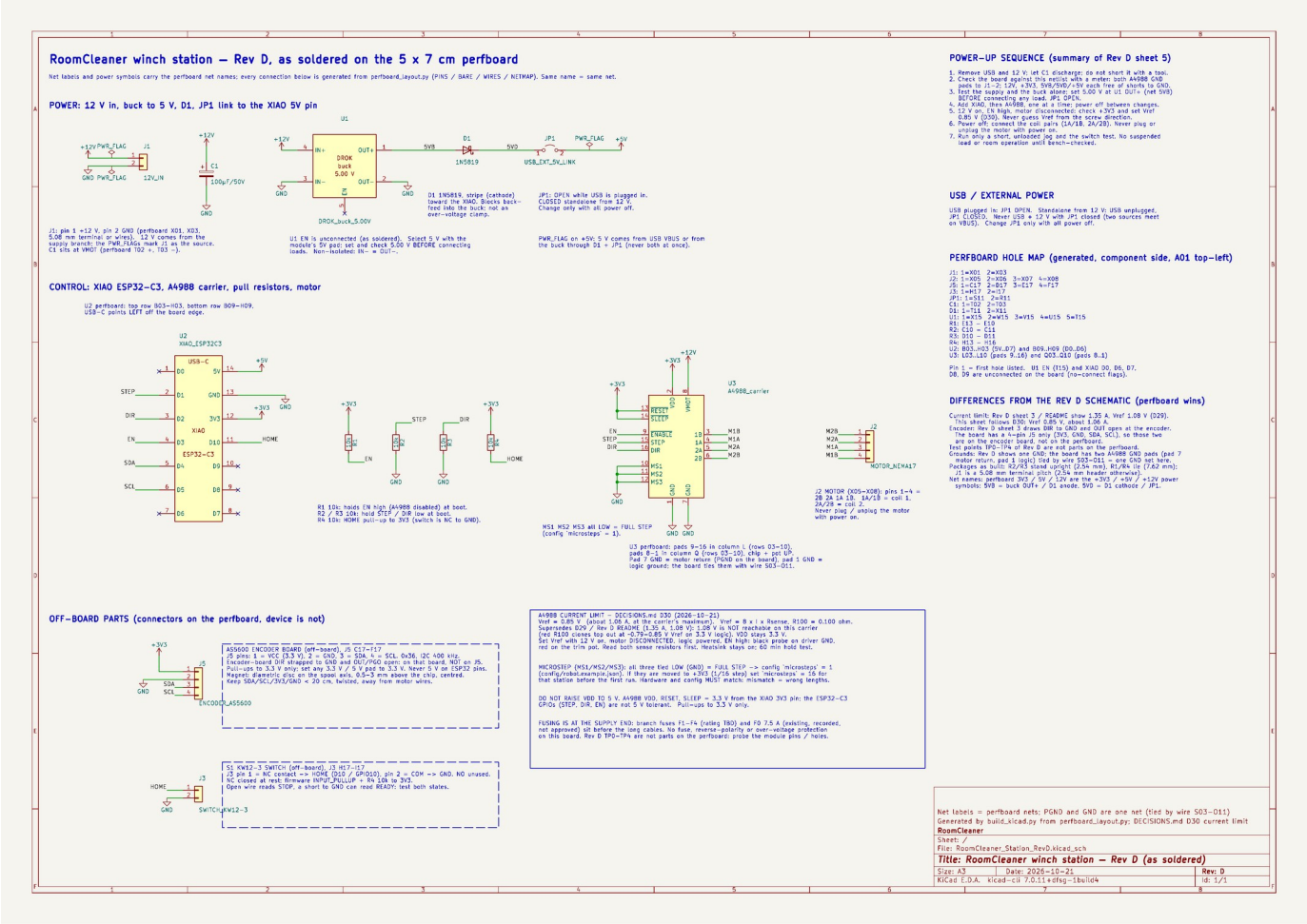
Fin: holes drilled from the first hole

My SolidWorks parts; blue and tan are cut faces, dark grey the bought bearing

Feature	As drawn	Fit
Gear bearings (687)	Ø14.075 pocket behind a Ø14.000 lip, 1.95 long, 8 T-slots	Snaps past the lip, +0.075 behind it
Gear pins	Printed Ø7.000 pin into a Ø7.05 socket	Line-to-line in the bearing; +0.05
Slide-in mount	0.175 per side; 30° dovetail flanks	Slip fit, 0.14–0.21 on the flanks
Spool on Ø5 D-shaft	Ø5.30 bore (10/21), Ø5.45 in v3; 0.5 mm flat	+0.30 to +0.45: Ø5.2 printed too tight
End-stop switch	Pocket 6.65 × 20.3 × 9 on a 1 mm ledge, lever window below	Press fit, no screws
Heat-set inserts	M3: Ø4.0 × 5 gear housing; M2.5: Ø3.45 × 4 encoder	Melt-in; hole deeper than the insert
Screw clearance	M3 Ø3.5 printed, Ø3.4 drilled; wall Ø3.97 (5/32 in)	+0.5 printed, +0.4 drilled
Motor pilot, Ø22 boss	Ø22.4 printed tab, Ø22.5 drilled fin	+0.4 / +0.5
Fin hole zero	Edges locate the bottom-left hole; drilled, it is the zero for the other 4	Pattern ±0.1, free of the ±0.5 sheared edges

- Printed holes come out small, so every printed fit carries an allowance; the drilled fin needs none
- PETG on a Creality K1, 0.4 mm nozzle, 0.2 mm layers; loads along the layers, mating faces off supports

Station electronics, Rev D

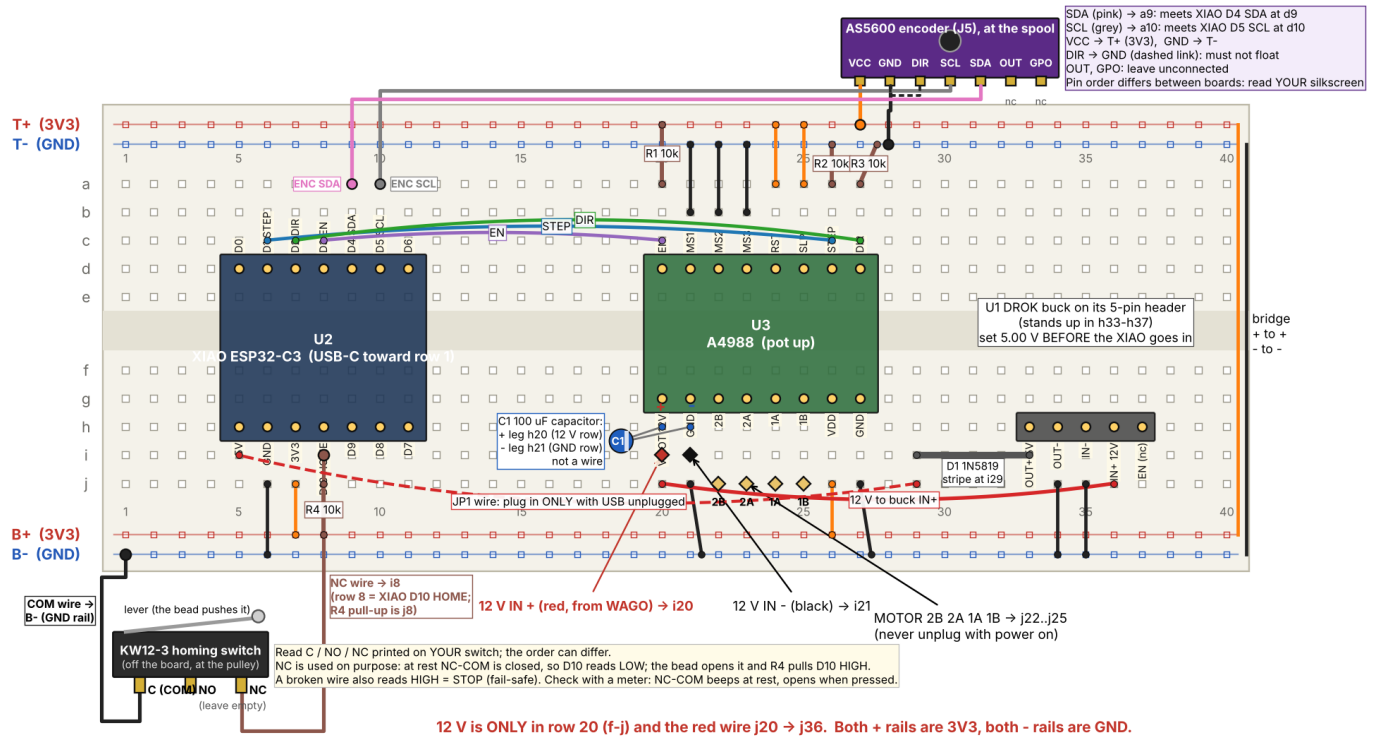


Station Rev D schematic

- One board per station: a Wi-Fi microcontroller, the A4988 driver, a 12 V to 5 V buck, the homing switch and the spool encoder
- Pull-up on ENABLE keeps the driver off at boot; pull-downs on STEP and DIR; a 100 µF reservoir at the driver supply
- Fuses at the source end of each 12 V branch; the buck is rated to 20 V in, so the supply stays at 12 V

Station breadboard layout

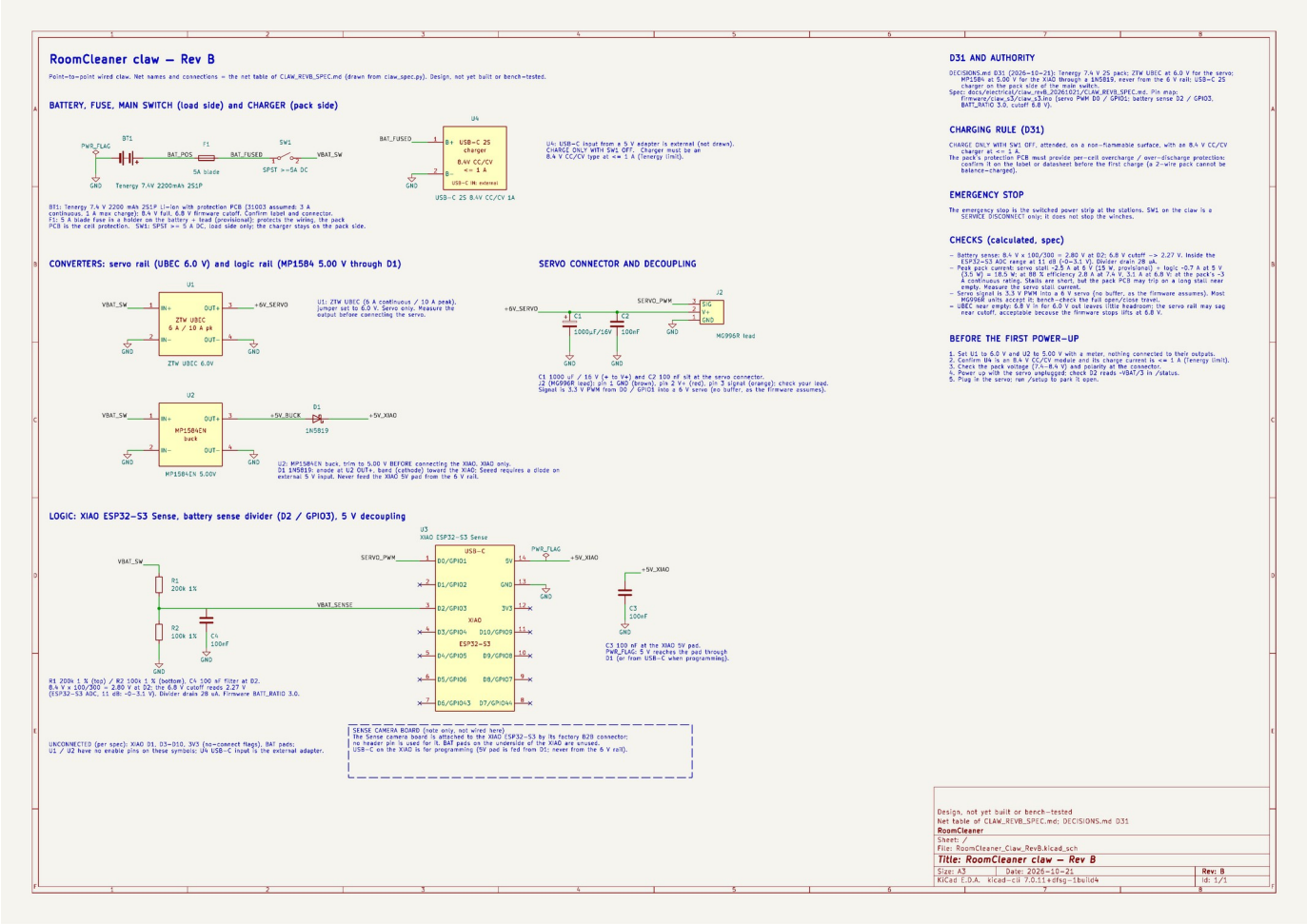
RoomCleaner winch station Rev D on a breadboard (full-size 830). Same nets as the perfboard and KiCad schematic.



- The bench build of Rev D on a mini breadboard: the layout was net-checked against the schematic (PASS) before power
- Each motor coil carries about 1 A through breadboard contacts: short leads, fully seated
- The soldered perfboard and the rail board follow the same net list

Station Rev D breadboard layout, drawn and net-checked

Claw electrical, Rev B



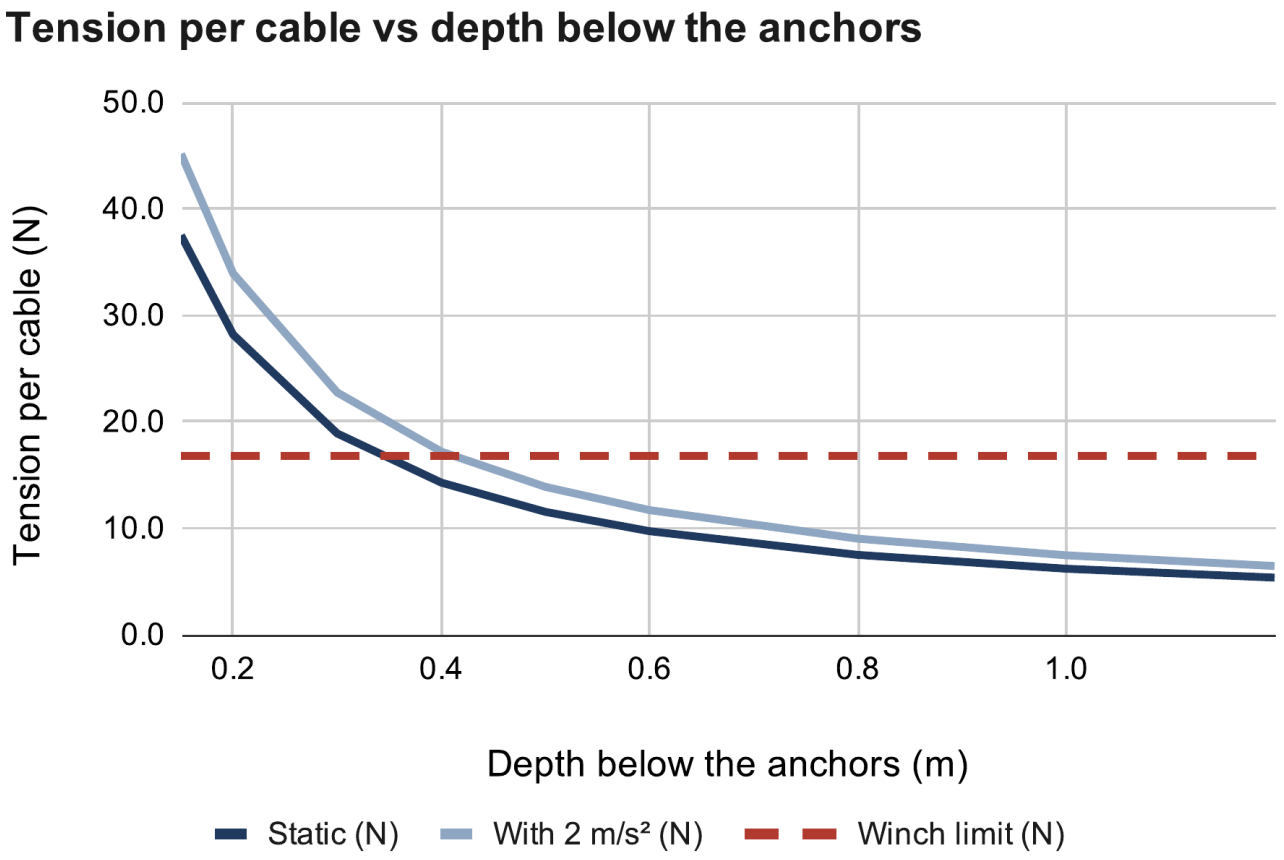
- Two-cell lithium-ion pack, 7.4 V; 5 A fuse for shorts
- 6 V servo rail and a separate 5 V logic rail; 200k/100k divider for battery sense
- Peak draw 2.8–3.1 A, at the pack's 3 A rating

Claw Rev B schematic

Acronyms

Term	Meaning
CAD	Computer-aided design
PETG	Polyethylene terephthalate glycol (the printed plastic)
TPU	Thermoplastic polyurethane (flexible printed plastic)
NEMA 17	National Electrical Manufacturers Association motor frame, 1.7 in square
CNC	Computer numerical control
ESP32	A Wi-Fi microcontroller family (Espressif)
A4988	Stepper-motor driver chip (Allegro)
YOLO	“You Only Look Once”: an object-detection model family
SF	Safety factor: allowable stress divided by actual stress

Calculations and sources



Calculated (calc sheet: Cable tension)

- All charts come from my engineering calculation sheet
- Incropera et al., Fundamentals of Heat and Mass Transfer, ch. 3
- Allegro MicroSystems, A4988 datasheet